University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

Operating System

Lab No 3

C Programming

Faculty of Information Technology

UCP Lahore Pakistan

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Topics to be covered

Basic C program coding

a. Data Type
b. Input/output
c. Control structures (If/else/switch)
d. Looping
e. Functions with/without parameters
f. Command Line Arguments
g. Dynamic memory allocation
h. Structures

Objectives:

- Students able to understand the concept of C Programing.

Pre-requisite:

- GCC installed in the system
- Visual studio code must be installed.
- Basic concept of C++

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Basic C programing

Data Type

Type	Size (Bytes)	Format Specifier
int	at least 2, usually 4	%d
char	1	%c
float	4	%f
double	8	%lf
short int	2 usually	%hd
unsigned int	at least 2, usually 4	%u
long int	at least 4, usually 8	%li
long long int	at least 8	%lli
unsigned long int	at least 4	%lu
unsigned long long int	at least 8	%llu
long double	at least 10, usually 12 or 16	%lf
signed char	1	%c
unsigned char	1	%c

Input/output

```
#include <stdio.h>
int main()
{


}
printf("Hello, World!");
return 0;
```

The #include <stdio.h> is a preprocessor command. This command tells compiler to include the contents of stdio.h (standard input and output) file in the program. The stdio.h file contains functions such as scanf () and printf () to take input and display output respectively. If you use printf () function without writing #include <stdio.h>, the program will not be compiled. The execution of a C program starts from the main () function. The printf () is a library function to send formatted output to the screen. In this program, the printf () displays Hello, World! Text on the screen. The return 0; statement is the "Exit status" of the program. In simple terms, program ends with this statement.

Task 1

Write a C program that asks the user to enter the radius of a circle. It calculates the area and circumference and displays both on screen (use float or double, and 3.14159 for π).
Area = π * radius * radius
Circumference = 2 * π * radius

Sample Output:

What is the radius of the circle? 7
The area of the circle is 153.94.
The circumference of the circle is 43.98.

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Control Structures

C if Statement

The syntax of the `if` statement in C programming is:

```
if (test expression)
 {
 // statements to be executed if the test expression is true
 }
```

How if statement works?

The `if` statement evaluates the test expression inside the parenthesis `()`.
If the test expression is evaluated to true, statements inside the body of `if` are executed.
If the test expression is evaluated to false, statements inside the body of `if` are not executed.

C if...else Statement

The `if` statement may have an optional `else` block. The syntax of the `if..else` statement is:

```
if (test expression) {
 // statements to be executed if the test expression is true
}
else {
 // statements to be executed if the test expression is fals e
}
```

How if...else statement works?

If the test expression is evaluated to true, statements inside the body of `if` are executed.

Statements inside the body of `else` are skipped from execution. If the test expression is

evaluated to false, statements inside the body of `else` are executed statements inside the body of `if` are skipped from execution.

Task 2

Prompt the user to enter the marks (out of 100) of a student. Determine if the marks are valid (between 0 and 100). If so, then determine the grade:

- A: 80 and above
- B: 70 to 79
- C: 60 to 69
- D: 50 to 59
- F: below 50

Hint: Marks below 0 or above 100 are invalid and should print an error message.

Sample Output:
Enter the marks: 74
Grade: B
Enter the marks: 120
Invalid marks!

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Looping

The syntax of the `for` loop is:

```
for (initializationStatement; testExpression; updateStatement)
{
  // statements inside the body of loop
}
```

How for loop works?

The initialization statement is executed only once. Then, the test expression is evaluated. If the test expression is evaluated to false, the `for` loop is terminated. However, if the test expression is evaluated to true, statements inside the body of `for` loop are executed, and the update expression is updated. Again the test expression is evaluated. This process goes on until the test expression is false. When the test expression is false, the loop terminates.

Task 03

Write a program to perform Selection Sort. The program shall take an array of 15 integers as input. If the user wants to enter fewer than 15 numbers, the user shall terminate input with -1 . The program shall then sort the sequence in **descending** order and print both the original and sorted sequence.

Sample Input:

Input Sequence: 8 21 4 17 9 30 2 -1

Sample Output:

The entered sequence is : 8 21 4 17 9 30 2
Updated sequence is: 30 21 17 9 8 4

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Functions

A function is a block of code that performs a specific task.

```
#include <stdio.h> void
functionName(){
}
Int main(){
functionName();
}
```

Task 4:

Write a void checkPalindromeNumber() function which takes an integer input from the user, checks whether it is a palindrome or not, and displays the result on the screen.

Sample Output:

Enter the number: 12321
Output: Palindrome

Enter the number: 4506
Output: Not Palindrome

Command Line Arguments

Command line argument is a parameter supplied to the program when it is invoked. Command line argument is an important concept in C programming. It is mostly used when you need to control your program from outside. Command line arguments are passed to the `main()` method.

```
int main (int argc , char ** argv)

#include <stdio.h>
int main(int argc, char
**argv){ printf("No of
arguments %d",argc);
 return 0;
```

Task 5:

Write a C program to receive two integers from the command line and display their sum, difference. If the number of arguments is incorrect, or if the second number is zero, print an appropriate error message.

Sample Output:
./a.out 20 4
Sum: 24
Difference: 16

Dynamic memory allocation

The name "malloc" stands for memory allocation.
The `malloc()` function reserves a block of memory of the specified number of bytes. And, it returns a pointer of `void` which can be casted into pointers of any form.

```
ptr = (castType*) malloc(size);

ptr=(int*) malloc(100 *sizeof(float));
```

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Task 6:
Write a C program to display the second largest number in an array. The user enters the number of elements first, and you must allocate memory at run time for the array using malloc(). Free the memory before exiting.
Sample Output:
Enter the number of elements: 6
Enter the array elements: 12 45 7 33 45 19
Output: 33

Task 7:
You are given a file "task8.txt" containing records of 10 books (Fig. 1). Write a C program to manage the library's book information using structures. The information of a book contains ID (e.g. Book01), Title (without spaces, e.g. DataStructures), Author (without spaces), Category (e.g. fiction/technical, without spaces), Edition (e.g. 1/2/3) and Price. All data will be saved to a file in the specific format shown in Fig. 1. The maximum number of books is 100. The program will prompt the user with a menu for different operations as shown below:

========== MENU ==========
1. Add a record
2. Search a record by ID
3. Show all records
4. Show books with price greater than a given limit (1000)
5. Save and exit

Task8.txt
Book01 DataStructures Weiss technical 3 1800
Book02 Godan Premchand fiction 1 650
Book03 OperatingSystems Silberschatz technical 9 2400
Book04 Aangan KhadijaMastoor fiction 2 800
Book05 CProgramming Kernighan technical 2 1500
Book06 Umrao Ruswa fiction 1 700

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REFERENCE LINK

HTTPS://WWW.PROGRAMIZ.COM/C-PROGRAMMING/C-FILE-INPUT-OUTPUT